

SIMATS ENGINEERING

ASSESSMENT TOOL 3: ASSIGNMENT 2

COURSE NAME: Programming in Java **COURSE CODE:** CSA0915- SLOT B
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ASSESSMENT TASKS:

A college maintains student records. Every student has a roll number, name, and department. A graduate student additionally has a research topic and guide name.

Tasks:

1. Create a Student class with a parameterized constructor to initialize roll number, name, and department.
2. Create a GraduateStudent class that inherits from Student.
3. Use the super() constructor to initialize the parent class attributes.
4. Display all student details including the research topic and guide name.

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

(Constructor Chaining)
5 marks

Q4 ✓
Bank Account and Savings
Account (Inheritance +
Constructor)
5 marks

Q5 ✓
Person, Faculty, and Professor
(Multilevel Inheritance +
Constructors)
5 marks

5/5 answered

```
1 class Student {
2     protected String rollNo;
3     protected String name;
4     protected String department;
5
6     public Student(String rollNo, String name, String department) {
7         this.rollNo = rollNo;
8         this.name = name;
9         this.department = department;
10    }
11
12    public void display() {
13        System.out.println("Roll Number: " + this.rollNo);
14        System.out.println("Name: " + this.name);
15        System.out.println("Department: " + this.department);
16    }
17
18    public static void main(String[] args) {
19        GraduateStudent gradStudent = new GraduateStudent(
20            "CS101",
21            "Alice Smith",
22            "Computer Science",
23            "Artificial Intelligence in Healthcare",
24            "Dr. John Doe"
25        );
26        gradStudent.display();
27    }
28 }
29
```

stdin input

Output

```
Roll Number: CS101
Name: Alice Smith
Department: Computer
Research Topic: Artif
Intelligence in Healt
Guide Name: Dr. John
```



A vehicle showroom maintains details of different vehicles. Every vehicle has a brand and model. Electric vehicles also have battery capacity and charging time.

Tasks:

1. Create a Vehicle class with both default and parameterized constructors.
2. Create an ElectricVehicle class that extends Vehicle.
3. Invoke the parent constructor using super().
4. Display all vehicle details.
5. Demonstrate constructor overloading

The screenshot displays the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo and the text "SIMATS Online Programming Lab" with a "JAVA PROGRAMMING" tab. On the left, a sidebar shows a list of questions, with Q4 and Q5 highlighted. The main area contains a code editor with the following Java code:

```
1 public class Vehicle {
2     protected String brand;
3     protected String model;
4
5     public Vehicle() {
6         this.brand = "Unknown Brand";
7         this.model = "Unknown Model";
8     }
9
10    public Vehicle(String brand, String model) {
11        this.brand = brand;
12        this.model = model;
13    }
14
15    public void display() {
16        System.out.println("Brand: " + this.brand);
17        System.out.println("Model: " + this.model);
18    }
19
20    public static void main(String[] args) {
21        ElectricVehicle eV1 = new ElectricVehicle();
22        eV1.display();
23
24        ElectricVehicle eV2 = new ElectricVehicle("Tesla", "Model S");
25        eV2.display();
26    }
27 }
28
```

On the right side, there is an "Input" section with "stdin input" and an "Output" section showing the results of the program execution:

```
Brand: Unknown Brand
Model: Unknown Model
Battery Capacity: 0 kWh
Charging Time: 0.0 hours
Brand: Tesla
Model: Model S
```

The bottom of the interface shows a Windows taskbar with various application icons and a system tray indicating 31°C and light rain.

A company stores employee information. Every employee has an employee ID, name, and basic salary. A manager also receives a monthly bonus.

Tasks:

1. Create an Employee class with a parameterized constructor.
2. Create a Manager class that inherits from Employee.
3. Use constructor chaining (super()) to initialize inherited data members.
4. Calculate and display the manager's total salary.
5. Create objects for at least three managers.



SIMATS Online Programming Lab

JAVA PROGRAMMING

Q4

Bank Account and Savings Account (Inheritance + Constructor)

5 marks

Q5

Person, Faculty, and Professor (Multilevel Inheritance + Constructors)

5 marks

5/5 answered

```

1 public class Employee {
2     protected int empId;
3     protected String name;
4     protected double basicSalary;
5
6     public Employee(int empId, String name, double basicSalary) {
7         this.empId = empId;
8         this.name = name;
9         this.basicSalary = basicSalary;
10    }
11
12    public void display() {
13        System.out.println("Employee ID: " + this.empId);
14        System.out.println("Name: " + this.name);
15        System.out.println("Basic Salary: " + this.basicSalary);
16    }
17
18    public static void main(String[] args) {
19        Manager m1 = new Manager(101, "Alice", 75000, 15000);
20        Manager m2 = new Manager(102, "Bob", 82000, 18000);
21        Manager m3 = new Manager(103, "Charlie", 90000, 20000);
22
23        m1.display();
24        System.out.println();
25        m2.display();
26        System.out.println();
27        m3.display();
28    }
29 }

```

stdin input

Output

```

Employee ID: 101
Name: Alice
Basic Salary: 75000.0
Monthly Bonus: 15000.0
Total Salary: 90000.0

```

A bank offers different account types. Every account has an account number, account holder name, and balance. A savings account also stores the annual interest rate.

Tasks:

1. Create an Account class with a parameterized constructor.
2. Create a SavingsAccount class that extends Account.
3. Initialize the inherited variables using super().
4. Calculate the annual interest earned.
5. Display the complete account details.



SIMATS Online Programming Lab

JAVA PROGRAMMING

Q4

Bank Account and Savings

Account (Inheritance + Constructor)

5 marks

Q5

Person, Faculty, and Professor

(Multilevel Inheritance + Constructors)

5 marks

5/5 answered

```

1 public class Account {
2     protected String accountNumber;
3     protected String accountHolderName;
4     protected double balance;
5
6     public Account(String accountNumber, String accountHolderName, double balance) {
7         this.accountNumber = accountNumber;
8         this.accountHolderName = accountHolderName;
9         this.balance = balance;
10    }
11
12    public void display() {
13        System.out.println("Account Number: " + this.accountNumber);
14        System.out.println("Account Holder: " + this.accountHolderName);
15        System.out.println("Balance: " + this.balance);
16    }
17
18    public static void main(String[] args) {
19        SavingsAccount savings = new SavingsAccount("SA10023", "John Doe", 50000.0, 4.5);
20        savings.display();
21    }
22 }
23
24 class SavingsAccount extends Account {
25     private double annualInterestRate;
26
27     public SavingsAccount(String accountNumber, String accountHolderName, double balance, double annualInterestRate) {
28         super(accountNumber, accountHolderName, balance);
29         this.annualInterestRate = annualInterestRate;
30     }
31
32     public void display() {
33         super.display();
34         System.out.println("Annual Interest Rate: " + this.annualInterestRate);
35         System.out.println("Annual Interest Earned: " + this.calculateInterest());
36     }
37
38     public double calculateInterest() {
39         return this.balance * this.annualInterestRate / 100;
40     }
41 }

```

stdin input

Output

```

Account Number: SA10023
Account Holder: John Doe
Balance: 50000.0
Annual Interest Rate: 4.5
Annual Interest Earned: 2250.0

```

31°C Mostly cloudy

Search

A university maintains faculty information. Every person has a name and age. A faculty member additionally has a faculty ID and department. A professor also has a specialization and years of experience.

Tasks:

1. Create a Person class with a parameterized constructor.
2. Create a Faculty class that inherits from Person.
3. Create a Professor class that inherits from Faculty.
4. Use constructor chaining (super()) in each class.
5. Display the complete details of the professor



2 marks

Q4

Bank Account and Savings
Account (Inheritance +
Constructor)
5 marks

Q5

Person, Faculty, and Professor
(Multilevel Inheritance +
Constructors)
5 marks


5/5 answered

```

1 public class Person {
2     protected String name;
3     protected int age;
4
5     public Person(String name, int age) {
6         this.name = name;
7         this.age = age;
8     }
9
10    public void display() {
11        System.out.println("Name: " + this.name);
12        System.out.println("Age: " + this.age);
13    }
14
15    public static void main(String[] args) {
16        Professor prof = new Professor(
17            "Dr. Emily Smith",
18            45,
19            "F1042",
20            "Computer Science",
21            "Machine Learning",
22            15
23        );
24        prof.display();
25    }
26 }
27
28 class Faculty extends Person {
29     protected String facultyId;

```

stdin input

Output

```

Name: Dr. Emily Smith
Age: 45
Faculty ID: F1042
Department: Computer Science
Specialization: Machine Learning
Years of Experience: 15

```